Le Qin

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Personal Profile

I have been a PhD student at The Hong Kong University of Science and Technology (Guangzhou) since 2022.08 under the supervision of **Prof. Jiayi Huang** and **Prof. Tushar Krishna**. My research interests primarily lie in the field of computer architecture and system, with a special focus on interconnection network and distributed machine learning systems. In addition, I am interested in hardware accelerators and machine learning algorithms.

Education

The Hong Kong University of Science and Technology (Guangzhou) — *Guangzhou, Guangdong, China*

Ph.D student in Microelectronics — *Aug 2022 - Present*

- Supervisor: **Prof. Jiayi Huang** Co-Supervisor: **Prof. Tushar Krishna**
- Research Area: Computer Architecture and System
- Research Topics: Collective Communication Optimization for Distributed Machine Learning, Efficient Scheduling for LLM Training and Inference

Nanjing University — *Nanjing, Jiangsu, China*

B.E. in VLSI Design and System Integration — *Sep 2018 - Jun 2022*

- Supervisor: **Prof. Jun Lin** and **Prof. Zhongfeng Wang**
- GPA: 4.32/5.00 (86.4/100)
- Courses: Digital IC Design, Analog IC Design, Embedded System, Digital Signal Processing, Digital Image and Computer Vision, Design of Deep Learning System Based on Hardware Acceleration of FPGA, GPU and Artificial Intelligence
- Honors & Prizes: People Scholarship (2019, 2020, 2021), Second Prize of National Undergraduate Electronics Design Contest in Jiangsu Division (2021), Outstanding Graduate in Nanjing University (2022)

Research Experience

Huang Lab at HKUST(GZ) — *Guangzhou, China*

Graduate Research Assistant — *Aug 2022 - Present*

- Explore optimization opportunities for collective communication in distributed deep learning
- Study software-hardware co-design for efficient DNN training/inference

Integrated Circuits and Intelligent Systems Lab (ICAIS) at NJU — *Nanjing, China*

Undergraduate Research Assistant — *Oct 2020 - Jun 2022*

- Study hardware acceleration for DNN/CNN models
- Study the design and implementation of hardware accelerators

Research Projects

XTree on EquiMesh: Topology and Algorithm Co-Design for Collective Communication — *HKUST(GZ)*

Second Author, Paper Published in **DATE 2026** — *Oct 2024 - Aug 2025*

- Analyze the All-Reduce communication bottleneck caused by non-uniform node degrees in 2D Mesh topologies, identifying the key insight that edge and corner nodes are the limiting factors for communication bandwidth utilization.
- Propose EquiMesh, a full-degree topology that achieves uniform node degrees by adding ring links along the edges without introducing routing complexity. Design the XTree topology-aware algorithm and the MirrorXTree mirroring mechanism to fully utilize bandwidth and enable efficient All-Reduce collective communication.
- Evaluation using an event-driven simulator demonstrates that, compared to state-of-the-art mesh topology-algorithm co-designs, EquiMesh achieves a 2× improvement in effective bandwidth for AllGather and a 1.2× improvement for AllReduce. Additionally, end-to-end LLM training performance is improved by 1.06× to 1.39×.

Optimizing All-to-All Collective Communication with Fault Tolerance on Torus Networks — *HKUST(GZ)*

First Author, Paper Published in **MICRO 2025**, Passing Artifact Evaluation with All Badges (Available, Evaluated, Reproduced) — *Dec 2023 - Aug 2024*

- Identify the growing All-to-All communication overhead in large-scale distributed deep learning computing, including Deep Learning Recommendation Model (DLRM) and Mixture-of-Experts (MoE) models.
- Propose twofold optimizations for fault-free All-to-All collective on torus network: HalfRing algorithm in each ring topology to construct shortest communication path and DimRotation scheduling for full bandwidth utilization across all the dimensions on N-D torus.
- Introduce fault-free All-to-All against link failures in two ways: FoldedRing algorithm facilitates fault-tolerant communication on a single ring and MATE scheduling accelerates communication in the faulty ring by utilizing available links of other dimensions.
- Conduct comprehensive experiments to prove the effectiveness of our methods using ASTRA-sim. The experiments include performance speedup, All-to-All bandwidth, dimension utilization, scalability study, and end-to-end training/inference speedup. Compared with fault-free baseline, our fault-free and fault-tolerant (with one link failure) scheme can achieve 2.28× and 1.37× speedup on average, respectively.

Communication Fusion for Hybrid Parallelism in Large Language Models *HKUST(GZ)*

First Author, Paper Published in **ISCA 2025** **++ ISCA 2025 Distinguished Artifact Award (3/570)** *Oct 2023 - Nov 2024*

- Scrutinize the communication patterns of each parallelism pattern and identify the redundant data movements problem in hybrid parallelism of distributed LLMs, which reveals the communication fusion optimizations for cascaded hybrid parallelism.
- Propose our communication fusion framework for hybrid parallel LLMs. The framework can fuse two adjacent collective communications and replace them with their subset or new collective. By chaining communication fusions of any cascaded hybrid parallelism pairs, our framework can effectively reduce the communication overhead of any complex hybrid parallelism.
- Conduct extensive evaluations on common LLM models by simulation with BookSim and SCALE-sim. The results show that our framework can bring 1.23×-7.06× network bandwidth speedup, and 1.16×-1.58× end-to-end performance speedup on average.

MoC-System: Efficient Fault Tolerance for Sparse Mixture-of-Experts Model Training *HKUST(GZ)*

Second Author, Paper Published in **ASPLOS 2025** *Mar 2024 - Oct 2024*

- Identify the inefficiency of traditional fault-tolerant checkpointing in sparse Mixture-of-Experts (MoE) model training, due to the explosion of expert parameters and distributed expert parallelism.
- Design the Mixture-of-Checkpoint (MoC) System with a novel Partial Experts Checkpointing (PEC) technique, which saves only a subset of experts during each checkpoint while preserving overall model accuracy.
- Implement fully sharded and two-level asynchronous checkpointing (snapshot + persist) with adaptive configuration and dynamic-K mechanisms, reducing both runtime overhead and storage load.
- Achieve up to 98.9% reduction in checkpoint overhead and an average 1.08% accuracy improvement on downstream tasks, with scalability demonstrated on both language and vision MoE models.

Shortcut-connected Expert Parallelism for Accelerating Mixture of Experts *HKUST(GZ)*

Third Author, Paper Published in **ICML 2025** **++ Adopted by Meituan LongCat-Flash 560B MoE Model** *Sep 2023 - May 2024*

- Reveal the communication bottleneck in expert parallelism for sparse Mixture-of-Experts (MoE) models, where All-to-All communication limits system scalability and efficiency.
- Propose ScMoE, a novel shortcut-connected MoE architecture with adaptive overlapping strategy, enabling full decoupling and parallelization of communication and computation between Transformer layers.
- Design and evaluate multiple shortcut configurations (Pos-1/2/3) and gating mechanisms (DirectAdd, CG-1, CG-2), identifying the optimal (Pos-2 + CG-1) configuration that balances accuracy and performance.
- Achieve up to 1.49× speedup in training and 1.82× in inference over top-2 MoE baseline, with comparable or superior model quality across vision and language models such as SwinV2-MoE, GPT2-MoE, and LLaMA2-MoE.

Network Acceleration of Input-Dynamic Communication for Large-Scale Deep Learning Recommendation Model *UCSB, UMN, HKUST(GZ), SamSung, UCSD, and HKUST*

Third Author, Paper Published in **ISCA 2025** *Aug 2022 - Aug 2023*

- Identify the communication bottleneck in the aggregation operation of DLRM embedding layer and the chance to leverage input reuse and output reuse to accelerate aggregation. Input reuse can avoid repeated injecting the same input to the network and output reuse alleviate network overhead by only transmitting reduction result to the GPU holding the output.
- Propose in-switch cache and in-network reduction to hardness both input and output reuse in input-dynamic communication operation like aggregation. In-network cache can cache data from messages that previously travel through a network switch and make the switch directly respond upon a cache hit. In-network reduction can hardness output reuse dynamically via a counter mechanism.
- Verify the effectiveness by simulation with gem5 Garnet simulator. The results show an average speedup of 3.12× to aggregation for a 64-GPU system and good scalability within low hardware overhead.

Classification Algorithm of Underwater Acoustic Recognition and Ship Radiated Noise *NJU*

Undergraduate Final Project *Dec 2021 - May 2022*

- Design the efficient data processing method and DNN model architecture dedicated for underwater acoustic recognition and classification of ship radiated noise. We explore typical characteristic extraction methods of ship radiated noise including MFCC, CQT, STFT and LOFAR. The choice of these pre-processing methods varies with classification and recognition tasks, and the results are used for the input of DNN training. We design ResNet_Deepship model based on ResNet50 with extra batch normalization and early fusion operation.
- Train the model with private datasets on 4× NVIDIA 3090 GPUs with 150 epoch. The accuracy on test datasets achieve 99.15% and 97.23% respectively for seven classification and nine classification of ship radiated noise. The accuracy of acoustic recognition is 74.72%. The whole method of data pre-processing and model training is published as a patent.

Algorithm-Architecture Co-Design for Accelerating Deep Reinforcement Learning *NJU*

Leader of National College Student Innovation and Entrepreneurship Project (Completion Score: Excellent) *Mar 2021 - Dec 2021*

- Accelerate Q-learning via algorithm optimization and chip architecture design. Traditional Q-learning suffers from slow computation of Q-value in the process of interacting with the environment. Additionally, the exploration is low-efficient with only one attempt at a time. We accelerate the process by allowing the agent to exploring all the possible directions at the same time instead of following the direction with highest possible reward. We design the algorithm and specific hardware architecture for acceleration.
- Evaluate the performance speedup by running maze tracing game. The speed of finding the optimal path in given random-generated maze gets over 3.21× speedup via computing Q-value in all four possible directions simultaneously. The method is published as a patent.

Publications

Conference Papers

XTree on EquiMesh: Topology and Algorithm Co-Design for Collective Communication

Junwei Cui, **Le Qin**, Weilin Cai, Jiayi Huang
Design, Automation Test in Europe Conference (DATE), 2026

Optimizing All-to-All Collective Communication with Fault Tolerance on Torus Networks
Le Qin, Junwei Cui, Weilin Cai, Meng Niu, Yan Yang, Jiayi Huang
Proceedings of the 58th International Symposium on Microarchitecture (MICRO), 2025

Chimera: Communication Fusion for Hybrid Parallelism in Large Language Models
Le Qin, Junwei Cui, Weilin Cai, Jiayi Huang
Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA), 2025

MoC-System: Efficient Fault Tolerance for Sparse Mixture-of-Experts Model Training
Weilin Cai, **Le Qin**, Jiayi Huang
Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Volume 2, 2025

Shortcut-connected Expert Parallelism for Accelerating Mixture-of-Experts
Weilin Cai, Juyong Jiang, **Le Qin**, Junwei Cui, Sunghun Kim, Jiayi Huang
International conference on machine learning (ICML). 2025

TRACI: Network Acceleration of Input-Dynamic Communication for Large-Scale Deep Learning Recommendation Model
Guyue Huang, Hao Li, **Le Qin**, Jiayi Huang, Yangwook Kang, Yufei Ding, Yuan Xie
Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA), 2025

Services

Program Committee

International Symposium on Computer Architecture (ISCA) Artifact Evaluation 2025
International Symposium on Microarchitecture (MICRO) Artifact Evaluation 2025
International Symposium on High-Performance Computer Architecture (HPCA) Artifact Evaluation 2026

Reviewer

International Conference on Computer Design (ICCD) Sub-Reviewer 2023-2025
IEEE International Symposium on Hardware Oriented Security and Trust (HOST) Sub-Reviewer 2026

Honors and Awards

2025	**MICRO 2025 Student Travel Grant Award**, MICRO’25	*Seoul, Korea*
2025	**ISCA 2025 Distinguished Artifact Award (3/570)**, ISCA’25	*Tokyo, Japan*
2025	**ISCA 2025 Student Travel Grant Award**, ISCA’25	*Tokyo, Japan*
2022-Present	**Full Postgraduate Scholarship**, HKUST(GZ)	*Guangzhou, China*
2022	**Outstanding Graduate**, Nanjing University	*Nanjing, China*
2021	**Second Prize of National Undergraduate Electronics Design Contest (Jiangsu Division)**, Ministry of Education, China	*Nanjing, China*
2019-2021	**People Scholarship**, Ministry of Education, China	*China*

Teaching

Teaching Assistant

MICS 6000W Parallel Computer Architecture HKUST(GZ), Fall 2023
MICS 6000L Advanced Computer Architecture HKUST(GZ), Spring 2024

Skills

Software C/C++, Python (PyTorch, Pandas, NumPy, Scipy, Matplotlib), CUDA Programming, Java
Hardware Verilog, System Verilog
Tools LaTeX, Git, MATLAB, Vivado Xilinx, Multisim, Altium Designer
Simulators gem5, ASTRA-sim, BookSim
Language English (Proficient), Chinese (Native)